

Named Entity Recognition

Depending on the language of your transcripts, you may be able to automatically tag dates and names of people, places and organisations, using the [Stanford Named Entity Recognizer \(NER\)](#).

The Stanford NER has recognizers for:

- Arabic
- Chinese
- English
- French
- German
- Hungarian
- Italian
- Spanish

Table 1: Example of named-entity tags

Enti- ties:	PER- SON	PER- SON			LOCA- TION					DATE
Words:	Presi- dent	Barack Obamawas	born	in	Hawaii.	He	was	elected	in	2008.

Several classifiers for English are included in the default installation. You can download classifiers for other languages from the [Stanford CoreNLP site](#).

Different classifiers include different possible entity labels:

- *3 class* : LOCATION, PERSON, ORGANIZATION
- *4 class* : LOCATION, PERSON, ORGANIZATION, MISC
- *7 class* : LOCATION, PERSON, ORGANIZATION, MONEY, PERCENT, DATE, TIME

The steps for Named Entity tagging your corpus are:

1. Install the Layer Manager
2. Configure a Named Entity layer

Install the Layer Manager

1. In LaBB-CAT, select the *layer managers* option on the menu at the top.
2. At the bottom, follow the link labelled: *List of layer managers that are not yet installed*.
3. Find the *StanfordNERecognizer* layer manager in the list, and press its *Install* button, then *Install* again.
You will see a configuration page with some information about the tagger, and an option to upload a tagger file, which we don't need to do.
4. Press *Configure*.
You will see a progress bar while the layer manager downloads the Stanford POS Tagger files.

Once it's finished, you'll see a further information page.

Create a Named Entity layer

Now the layer manager is installed, we need to create a layer that is configured to use it to tag words that are names of entities...

- (1) Select *word layers* on the menu at the top.
- (2) You will see a list of word tag layers that have already been configured. The column headings at the top are also a form for creating a new layer, so we'll fill in that form now.
- (3) Fill in the following details on the form at the top:
 - *Layer ID*: `namedEntity` ¹
 - *Type*: *Text*
 - *Alignment*: *None*

 - *Manager*: *Stanford Named Entity Recognizer*
 - *Generate*: *Always*
 - *Project*: This can be left as the default value, unless you want to add the layer to a category of your choice.
- (4) Press *New*
You will see the layer configuration form. Mostly you can leave the default values as they are.
- (5) By default, the *Classifier to use* options are all for English data; you can choose whether 3, 4, or 7 types of entities are tagged.

¹There may already a layer called `entity` in your LaBB-CAT configuration under *phrase layers*, so if you want to call this layer `entity`, you will have to delete the phrase layer first.

i Installing another Classifier

If your data is not in English, you can download a recognizer for your language from the [Stanford CoreNLP website](#)

To install a classifier:

1. Download the corresponding files from the CoreNLP website
 2. In the layer configuration form, set *Classifier to use* to [other classifier]
You will be asked to select a file.
 3. Select the file you just downloaded.
After a short delay while the file is uploaded, you'll see a message saying "Classifier(s) installed."
 4. In the layer configuration form, you can now set *Classifier to use* to the classifier you just installed.
- (6) Press *Set Parameters*.
- (7) Now press *Regenerate* to run the Named Entity Recognizer on your whole corpus.
You will see a progress bar while the transcripts are being tagged.
- (8) Once it's complete, select the *transcripts* option on the menu, and click the first transcript in the list.
- (9) Tick the new *namedEntity* layer to display the tags.
You will see that some words have a tag above them - these identify the type of named entity found by the classifier.

AP511_MikeThorpe

ah w~ . well I have a - m~ fairly vivid recollection um of
DATE
the first one being September fourth . where I had arrived
LOCATION LOCATION
ah in Hornby . so ah coming from Greendale it

Figure 1: Named Entity tags

Editing tags

The NER tags can be manually corrected if required.

To change the label of a tag on the transcript page:

1. Hover the mouse over the tag you want to change.
An 'edit' button will appear to the right.

hquake hit . um - she most certai
PERSON 
come from Napier . I've ah :

2. Press the edit button.
The tag will become a text box you can type in.

hquake hit . um - she most certai
LOCAT 
come from Napier . I've ah :

3. Change the tag's label as desired.
4. Press then save button  to save your changes.
You will see a message asking "Are you sure you want to save this tag?"
5. Press OK to confirm your change.

To delete a tag, follow the steps to change the label, and set the label to be blank before saving.

To add a new tag for a word that the NER has not recognized:

1. Hover the mouse in the space above the word you want tag, where the tag would be if it had been recognized.

An 'add tag' button will appear.

' the doorway a~ about six



because Victoria Street was

2. Press the 'add tag' button.
An empty text box will appear.

' the doorway a~ about six



because Victoria Street v

3. Type in the desired label.

e doorway a~ about six of us I'm not



use Victoria

Street was so badly a
Save changes to tag

DATE

4. Press the save button  to save your changes.
You will see a message asking "Are you sure you want to create this tag?"
5. Press *OK* to confirm your change.

If you manually add or edit Named Entity tags, and then run the StanforNER again, the manual changes will be preserved, although the tagger will re-create deleted entity tags.